A380056

Exponential coefficients and congruences modulo four and five

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AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

If $\sum a_n x^n/n! = (e^x - 1)/\cos(2x)$, then $a_n \equiv 1 \pmod 4$ for $n \geq 1$ and $a_n \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5$ for all $n \geq 0$. Thus $5 \mid a_n$ exactly when $4 \mid n$. Integer binomial convolution constructs the sequence and proves its literal rational exponential-generating-function identity.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A380056 [1], retrieved on 24 October 2026. Its initial sequence index is 1. We use the auxiliary value $a_0 = 0$ and distinguish the exponential coefficients $a_n = n![x^n]A$ from ordinary rational coefficients.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x + O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U \circ V = U(V(x))$ only when $V(0) = 0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1 \rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

2 A380056: exponential coefficients and binomial convolution

Here the source is the exponential generating function [1]

$$A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!} = \frac{e^x - 1}{\cos(2x)}, \qquad a_0 = 0. \tag{1}$$

The integer sequence must be distinguished from the ordinary rational coefficients of this formal series. We first establish that distinction and the source equation exactly.

For two sequences over a commutative ring, define binomial convolution by

$$(f \star g)_n = \sum_{k=0}^{n} \binom{n}{k} f_k g_{n-k}.$$

Write $\delta = (1, 0, 0, \ldots)$ and $d = (0, 1, 1, \ldots)$, and define

$$c_{2j} = (-4)^j, \qquad c_{2j+1} = 0.$$

Lemma 2.1 (Integral source construction)**.** *There is a unique integer sequence a satisfying $a \star c = d$. Its exponential generating function is exactly (1).*

Proof. Since $c_0 = 1$, set $a_0 = 0$ and, for $n > 0$, set

$$a_n = 1 - \sum_{k=0}^{n-1} \binom{n}{k} a_k c_{n-k}.$$

This determines integers without division and gives the stated convolution identity. Induction proves uniqueness over any commutative coefficient ring.

Over $\mathbb{Q}$, let $\mathcal{E}(f) = \sum_{n \geq 0} f_n x^n / n!$. The identity $\binom{n}{k} k!(n-k)! = n!$ gives $\mathcal{E}(f \star g) = \mathcal{E}(f)\mathcal{E}(g)$ coefficientwise. Also $\mathcal{E}(c) = \cos(2x)$ and $\mathcal{E}(d) = e^x - 1$. Thus $\mathcal{E}(a)\cos(2x) = e^x - 1$. The cosine has constant coefficient one and is a multiplicative unit, so its inverse gives (1), uniquely in $\mathbb{Q}[[x]]$. By construction $n![x^n]\mathcal{E}(a) = a_n$. □

Theorem 2.2 (The two A380056 conjectures)**.** *For every $n \geq 1$, $a_n \equiv 1 \pmod 4$. For every $n \geq 0$,*

$$a_n \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5. \tag{2}$$

Consequently the residues modulo five at index classes $0, 1, 2, 3$ modulo four are respectively $0, 1, 1, 3$, and

$$5 \mid a_n \quad \Longleftrightarrow \quad 4 \mid n.$$

In particular $5 \mid a_{4n}$ for every positive n.

Proof. All positive-index coefficients of c vanish modulo four, so $c = \delta$ in binomial convolution over $\mathbb{Z}/4\mathbb{Z}$. Reduction of $a \star c = d$ therefore gives $a = d$, proving the first assertion.

For the second, work with sequences over $\mathbb{F}_5$ and put $E_s(n) = s^n$, with $0^0 = 1$. The binomial theorem gives

$$E_s \star E_t = E_{s+t}.$$

These are divided-power coefficient sequences. No factorial denominators or ordinary exponential series over $\mathbb{F}_5$ are needed. Since $-4 = 1$ in $\mathbb{F}_5$, the sequence c is one at even indices and zero at odd indices, and hence

$$c = 3(E_1 + E_4), \qquad d = E_1 - E_0.$$

For $b = 2E_2 + 3E_4$, bilinearity gives

$$\begin{aligned} b \star c &= 6E_3 + 6E_1 + 9E_0 + 9E_3 \\ &= E_1 - E_0 = d \quad \text{over } \mathbb{F}_5. \end{aligned}$$

The recurrence in Lemma 2.1 remains uniquely solvable over $\mathbb{F}_5$, so $a = b$ after reduction, proving (2). Both 2^4 and 4^4 equal one modulo five. The four residues follow by evaluating the four index classes, and only class zero gives zero. □

The endpoints in `Round4.ExpCosine` prove the integer convolution recurrence, its uniqueness, the factorial coefficient interpretation, the literal quotient in (1), uniqueness among rational formal series, both conjectures, and the stronger complete classification modulo five. The source bridge uses Mathlib's formal cosine series rescaled by two.

3 Separate Lean proof and checked scope

The entry file is `lean/Entries/A380056.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A380056`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `integer_egf_definition`.
- `source_egf`.
- `literal_source_equation`.
- `literal_source_unique`.
- `integer_coefficients`.
- `first_source_conjecture`.
- `second_source_conjecture`.
- `exact_mod_five_pattern`.
- `exact_divisibility_by_five`.
- `mod_five_exponential_formula`.
- `mod_five_period`.

The main implementation module is `Round4.ExpCosine`. The catalogue records the full original endpoint names and the new entry theorems. The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The collection report records source recompilation and transitive axiom checks. Accepted axioms are propositional extensionality, classical choice, and quotient soundness. All index conclusions above are unbounded; exact finite coefficient checks in the package provide separate corroboration.

Limits and status. These are exponential coefficients $n![x^n]A$, not ordinary rational coefficients. The literal rational EGF and its uniqueness are formally proved. Modular convolution does not divide by factorials in finite characteristic. The manuscript consolidates completed results; it does not assert global novelty, exclusive priority, peer review, OEIS acceptance, or a new submission. Standard algebraic ingredients are not claimed as new discoveries.

References

[1] Paul D. Hanna, *OEIS A380056*, 28 January 2025. `https://oeis.org/A380056`. Retrieved 24 October 2026.